Information Matters: Sibling Influence on Study Choice

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Abstract

This paper studies spillovers from older to younger siblings' study choices. I use Dutch admission lotteries in medical schools to directly test the role of the preference and information mechanisms. I find that older siblings influence younger siblings' field of study choice but not university enrollment decisions. I show that 70% of sibling spillovers are due to information sharing, and the remaining 30% are due to common preferences.

Keywords: study choice, siblings, instrumental variable, the Netherlands

JEL Codes: I23, I24

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1 Introduction

How do people make decisions in their early life that will affect their adult outcomes? Who helps them to make these decisions? A field of study choice is one of such pivotal decisions that a person makes in life. This field of study choice is consequential in determining what career path to pursue. Many papers find a wide range of returns to different studies (Altonji et al., 2012, 2016). Usually, more math-oriented fields of study tend to produce higher returns than social sciences and humanities (Altonji et al., 2012). Further, certain fields, e.g. medicine and STEM, often have an insufficient number of students, though are seen as socially beneficial. Hence, a decision on what field to specialize in is crucial in determining labor market outcomes and affecting societal benefits. Less is known about how a person chooses a field of study and who can influence a person's study choice. Family networks and peers are found to influence important life choice decisions. Arguably, the closest peers to a person at a young age are one's siblings. Siblings can directly influence each other's behavior by simply observing what decisions they make in everyday life. Understanding how siblings – one of the most important peers – can influence each other's educational choices – one of the most important life decisions – is a relevant empirical question.

This paper studies how older siblings influence the study choice decisions of their younger siblings. Identifying peer effects is a challenging problem due to correlated effects and reflection problem (Manski, 1993). I overcome this problem by exploiting a unique setting where admission to medical schools is determined by a lottery. I use administrative data on a universe of students who have ever participated in admission lotteries. My identification strategy relies on the fact that admission to medical schools is random, hence, any differences between the study choice of younger siblings, whose older siblings are lottery winners or lottery losers can be attributed to the influence of the older sibling's enrollment.

I find that older siblings have a significant and economically meaningful influence on younger siblings' field of study choice but do not influence university enrollment decisions. I find that enrollment of an older sibling in a lottery program increases the probability of younger siblings being enrolled in a program to which an older sibling has *applied* by 3.2 percentage points. Usually, siblings choose the same program

3.2% of the time, hence, this effect translates to a 98.7% increase relative to the mean. A younger sibling is only 1.2 percentage points more likely to be enrolled in a program where an older sibling is *enrolled* (21.6%). The results provide evidence that older siblings mostly act as information providers to their younger siblings, while siblings' study preferences are correlated only to a minor extent. Hence, the main contribution of this paper is a direct test of two mechanisms in sibling spillovers: common preferences and information sharing.

This paper relates to a limited but growing amount of papers studying siblings spillovers in universities (Altmejd et al., 2021; Aguirrea and Matta, 2021; Goodman et al., 2015) and secondary schools (Dahl et al., 2021; de Gendre, 2021; Gurantz et al., 2020; Dustan, 2018; Joensen and Nielsen, 2018). The closest in spirit to this study is a paper by Altmejd et al. (2021). They study sibling spillover effects in college and major choice across four countries. They find that younger siblings follow their older siblings to the same college and the same college-major. In line with these findings, Aguirrea and Matta (2021) find strong sibling spillover effects on university enrollment but not on the field of study choice. To explain the mechanisms of sibling spillovers, previous papers rely on analyzing solely heterogeneous treatment effects. However, in the paper, I use the unique institutional setting of admission lotteries in the Netherlands to test directly which mechanism dominates.

A simple model exploiting admission lotteries predicts competing effects of an older sibling's study choice on a younger sibling's study choice. In this model, there are two types of programs: lottery programs that use a lottery to admit students, and programs without a lottery. The highly stylized model makes key comparative static predictions very clear. First, under the pure preference mechanism, a model predicts a *zero* effect of an older sibling's study choice on whether a younger sibling would be enrolled in the program which an older sibling has *applied*, while it predicts a *positive* effect on whether a younger sibling would be enrolled in the program which an older sibling is *enrolled*. Second, under the pure information mechanism, a model makes opposite predictions: a *positive* effect and a *zero* effect, respectively.

To explore whether my results can be explained by other channels, i.e. imitation or sibling rivalry, I show the results of sibling spillovers on university enrollment as well as heterogeneous effects. There are no sibling spillover effects on enrollment in a university or enrollment in the same university. Besides, sibling spillover effects

do not differ across older siblings of different abilities. These findings speak against an imitation mechanism since younger do not simply follow their older siblings to a university and they do not rely more heavily on their more academically successful siblings. I find little heterogeneity by the sex composition of siblings, indicating that sibling rivalry is not a likely factor in determining study choice decisions. I find that sibling spillover effects increase with the age difference. This indicates that more academically experienced older siblings provide more information about study programs, i.e. not only information about the application process but about the study experience. The results are robust and stay virtually unchanged using different sample definitions.

Understanding the mechanisms of sibling spillover matters for educational reforms. The preference mechanism and the information mechanism have two different policy implications. If the preference mechanism is prevailing in sibling spillovers, a policy maker cannot directly influence the decision of choosing a specific field. Preferences are formed before the application process, which makes it hard to change one's decisions. If the information provision mechanism is prevailing, programs and campaigns aimed at increasing enrollment in specific fields of study would have multiplicative effects. From a policy perspective, the results of the paper imply that additional information about the program can help in attracting more students into these fields. Particularly, such informational programs would have a greater influence on students from families with lower income and families with a migrant background.

2 Institutional background

In the Netherlands, students who complete the academic track in secondary school can enroll in any Dutch university. The majority of university programs have to accept all students, while some programs have quotas with a limited number of places, e.g. medical schools.

Until 1999, medical schools used a nationwide centralized lottery to admit students. The lottery was introduced as a response to the increasing number of applicants to medical schools which exceeded the number of available places. The Ministry of Education determines quotas. If the number of applicants exceeds the

number of available places, a lottery determines who is admitted. The lottery is weighted, so students with a higher GPA have a higher probability of winning. A GPA is determined based on the grades from nationwide and externally graded high school exams, where six indicates a pass. Table 1 shows the weights and shares of groups of students in different lottery categories. For a more detailed description of Dutch admission lotteries and the educational system, read Ketel et al. (2016).

Table 1: Lottery categories

Category	GPA	Share	Weight
A	$GPA \geq 8.5$	0.02	2.00
B	$8.0 \leq GPA < 8.5$	0.05	1.50
C	$7.5 \leq GPA < 8.0$	0.09	1.25
D	$7.0 \leq GPA < 7.5$	0.21	1.00
E	$6.5 \leq GPA < 7.0$	0.22	0.80
F	$GPA < 6.5$	0.30	0.67
Other	–	0.11	1.00

Notes: A GPA is a grade point average on the final exams in high school. Share is the share of applicants in different categories that applied for the lotteries in the years 1988-1999. Weight indicates the relative probability of being admitted. The category "other" refers to students who did not participate in the nationwide high school exams, such as foreign students. This category is excluded from the analysis.

3 Predictions

There are multiple reasons why siblings follow similar educational pathways. First, siblings can have common educational preferences. Siblings are exposed to a similar environment, e.g. family, and they might attend the same school. As a result of such a close connection, it is quite common for siblings to choose a similar field of study. This mechanism is called the common preference mechanism.

Second, an older sibling can provide valuable information about a field she is enrolled in. For example, an older sibling can inform a younger sibling about specific courses, the quality of peers, and the level of a program which can be helpful for a younger sibling to choose a field. This type of information is hard to observe

without studying in a program. Older siblings lower the costs of obtaining such information and make study options more salient to their younger siblings. Hence, by enrolling in a specific program an older sibling can provide information to a younger sibling who would not have received it otherwise. This mechanism is called the information–sharing mechanism.

To distinguish which mechanism is prevailing, I can use admission lotteries to medical schools. Admission lotteries allow us to estimate two treatment effects:

1. The effect of an older sibling's enrollment in a lottery program on whether a younger sibling is enrolled in the program where an older sibling has *applied*:

$$\Delta_{LP} = \mathbb{E}[LotteryProg_{YS}|Winner_{OS}] - \mathbb{E}[LotteryProg_{YS}|Loser_{OS}]$$

2. The effect of an older sibling's enrollment in a lottery program on whether a younger sibling is enrolled in the program where an older sibling is *enrolled*:

$$\Delta_{SP} = \mathbb{E}[SameProg_{YS}|Winner_{OS}] - \mathbb{E}[SameProg_{YS}|Loser_{OS}]$$

If younger siblings are enrolled in a program where their older siblings *applied*, it indicates common educational preferences. In other words, siblings have similar preferences about what to study, hence, it matters whether younger siblings follow their older sibling's application decisions. If younger siblings are enrolled in the program where their older siblings are *enrolled*, it indicates that younger siblings follow the educational pathways of their older siblings.

Let us analyze a stylized example of a study choice model to predict these two treatment effects. Assume there are two study programs: medicine and economics. Medicine is a program that uses a lottery to admit students, i.e. a lottery program. There is a 50% probability of winning a lottery. Assume all lottery winners comply with the outcome of a lottery, i.e. full compliance. Economics is a program without a lottery, i.e. all students who apply are accepted.

First, let us consider a case with the pure preference mechanism in Figure 1. In this case, siblings have common educational preferences. Since all older siblings in the sample applied to a lottery, they revealed preferred medicine as their study choice. In that case, the study preferences of younger siblings are identical among older siblings who are lottery winners and lottery losers. Hence, when the preference

Figure 1: A model of study choice with the pure preference mechanism

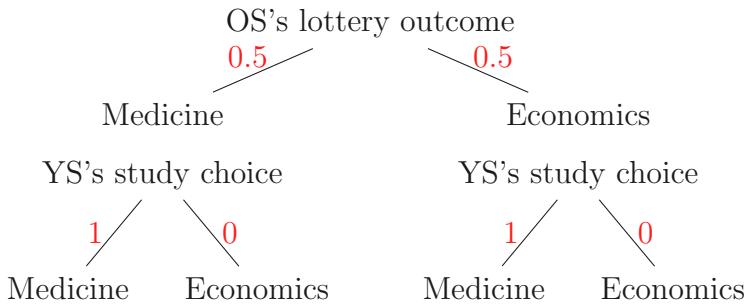
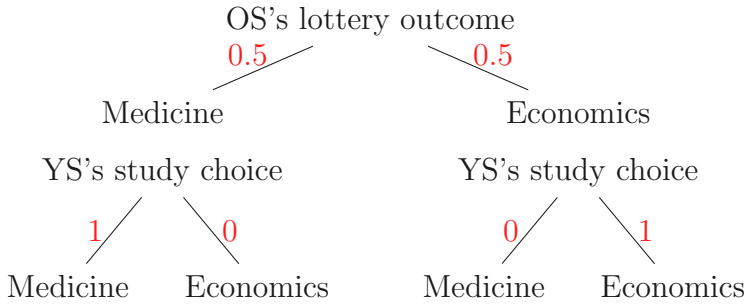


Figure 2: A model of study choice with the pure information mechanism



effect is prevailing, the effect of an older sibling's study choice on whether a younger sibling is enrolled in the program where an older sibling has *applied* is predicted to be *zero*, while the effect on enrollment in the program where an older sibling is *enrolled* is predicted to be *positive*.

Second, let us consider a case with the pure information mechanism in Figure 2. In this situation, older siblings provide information about a study program they are actually enrolled in. As a result, better-informed younger siblings are enrolling in the programs their older siblings are enrolled. Hence, when the information effect is prevailing, the effect of an older sibling's study choice on whether a younger sibling is enrolled in the program where an older sibling has *applied* is predicted to be *positive*, while the effect on enrollment in the program where an older sibling is *enrolled* is predicted to be *zero*.

The summary of model predictions is presented in Table 2. The magnitude of treatment effects can inform which mechanism dominates. As both mechanisms give distinct magnitudes of treatment effects, it is possible to empirically test which

mechanism is prevailing in the sample. These predictions are the main contribution to the literature as I can test directly the role of different mechanisms proposed in previous papers using a unique setting with admission lotteries.

Table 2: Predicted treatment effects of an older sibling's study choice on a younger sibling's study choice

Outcome	Common preferences	Information–sharing
Δ_{LP}	0	+
Δ_{SP}	+	0

4 Data

4.1 Data source and sample

I use administrative data from the Statistics Netherlands (CBS) which allow linking all individuals to their parents and siblings. The registry contains information on different demographic and educational outcomes, i.e. age, sex, immigration background, university enrollment, a field of study, and lottery outcomes. I observe lottery data on individuals who applied to medical schools between 1987 and 2004. For lottery participants, I observe the year when they participated in a lottery, lottery category, and lottery outcome.

To ensure that I observe first-year applicants, I exclude applicants who participated in the lottery in 1987 since there is no information about lottery participation in previous years. Since the lottery system was gradually abandoned from 1999 onward, I exclude lottery participants applying for the first time after this year. Applicants from category A are also excluded since almost all of them are admitted to a medical school. Further, I exclude applicants older than 20 at the moment they first participated in a lottery to exclude individuals who did not follow the usual educational track. Finally, I exclude applicants admitted via ministers' places since they do not participate in a lottery.

A unit of the analysis is a sibling pair. Hence, for every lottery participant, I link their younger siblings using a registry of their parents. I restrict the sample to

lottery applicants who have at least one younger (half)–sibling. I include siblings who are born at least one year and not more than twenty years apart. In case there are several older siblings in a family participating in a lottery, I take the oldest older sibling to determine the treatment status of a younger sibling. The results are identical when I use the youngest older sibling, when I include families with only one older sibling participating in a lottery, or when I restrict the sample only to two siblings families.

4.2 Descriptive statistics

Table 3 shows the balancing table of pre–determined characteristics of older and younger siblings by the outcome of the first lottery of older siblings. The table presents sample means, and p–values obtained from regressing personal characteristics on an indicator function for an older sibling winning the first lottery, including fixed effects for the program, lottery category, year of the first lottery participation, and their interactions. Around 60% of older siblings are female, 7% have a non–western background, and the average age at the first lottery is 18.8. The average number of siblings in a family is 3. Personal characteristics of older siblings are well balanced as indicated by p–values.

Table 3 also represents the personal characteristics of younger siblings. Around 50% of the younger siblings are female, 8% have a non-western background, and the average age difference between younger and older siblings is 4.5 years. Around 50% of sibling pairs are same sex pairs.

Table 4 presents descriptive statistics of enrollment of older and younger siblings. Almost all older siblings are enrolled in a university: 99% among winners and 95% among losers. Winning a lottery is associated with a 50 percentage points increase in enrollment in a lottery program for older siblings. For younger siblings, around 82% are enrolled in a university, while 9% are enrolled in a lottery program.

5 Empirical approach

In this section, I present my identification strategy to estimate sibling spillovers and derive testable predictions from a stylized model.

Table 3: Balancing of personal characteristics of older and younger siblings by the outcome of the first lottery of older siblings

Outcome	Lottery winners	Lottery losers	p-value
Older siblings			
Female	0.59	0.59	0.62
Non-western immigrant	0.07	0.07	0.39
Age at first application	18.82	18.86	0.74
Family size	2.97	2.97	0.42
N	8029	10002	
Younger siblings			
Female	0.49	0.48	0.02
Non-western immigrant	0.08	0.09	0.13
Age difference in years	4.50	4.43	0.35
Same sex pairs	0.49	0.50	0.21
N	12294	15348	

Note: The p–values in the final column are from the regressing personal characteristics on an indicator function for an older sibling winning the first lottery, including fixed effects for program, lottery category, year of the first lottery participation, and their interactions.

Table 4: Enrollment outcomes of older and younger siblings by the outcome of the first lottery of older siblings

Outcome	Lottery winners	Lottery losers
Older siblings		
Enrolled in a university	0.99	0.95
Enrolled in a lottery program	0.95	0.44
N	8029	10002
Younger siblings		
Enrolled in a university	0.83	0.81
Enrolled in a lottery program	0.10	0.08
Enrolled in the same university	0.22	0.21
Enrolled in the same program	0.07	0.06
N	12294	15348

Note: Since the lottery is weighted, the observed differences between lottery winners and losers in this table cannot be interpreted as causal.

I estimate the effect of an older sibling's study choice on a younger sibling's study choice, using the following model:

$$Y_i = \delta D_i + X_i\beta + \alpha_i + U_i \tag{1}$$

where Y_i indicates if in a sibling-pair i a younger sibling 1) is enrolled in the program where an older sibling has *applied* and 2) is enrolled in the program where an older sibling is *enrolled*; D_i indicates whether an older sibling is enrolled in a lottery program; X_i includes indicators for siblings' sex and non-western origin, age of an older sibling at the first lottery; α_i are the program, lottery category, year of the first lottery participation fixed effects, and their interactions.

The effect of an older sibling's study choice on a younger sibling's study choice is captured by δ. A naive OLS estimate of δ is biased in the presence of endogeneity. The compliance rate with the result of the lottery is imperfect. Not all winners enroll in medical school, while some losers reapply in subsequent years. Hence, I use instrument D_i with the result of the first lottery LR_{1i} in the following model:

$$D_i = \lambda LR_{1i} + X_i\theta + \kappa_i + V_i \tag{2}$$

Since the program admission is random conditional on the program, lottery category, year of the first lottery participation fixed effects, and their interactions, the zero mean conditional independence assumption holds for the first lottery: $\mathbb{E}[V_i|X_i, \kappa_i, LR_{1i}] = \mathbb{E}[V_i|X_i, \kappa_i]$. λ indicates the compliance rate, i.e. the difference in enrollment rates between winners and losers. λ describes the fraction of compilers in the sample for whom enrollment in a medical school is determined by the result of the first lottery. Hence, I estimate the effect of an older sibling's enrollment in a lottery program on the younger sibling's probability to be enrolled in a program where an older sibling has *applied* or is *enrolled*. Admission lotteries help me to overcome correlated effects and the reflection problem as the variation in the older sibling's study choice is exogenously determined by a lottery and is not influenced by the younger sibling's study choice. I cluster standard errors at the family level, as treatment is assigned at this level and there can be multiple siblings within a family.

6 Results

6.1 Main results

This section presents main results of how older siblings influence younger siblings' study choice decisions, i.e. field of study and enrollment to a university.

Siblings spillovers on field of study choice

Table 5 shows reduced form estimates of sibling spillovers on the field of study choice. Reduced form estimates capture the increased probability of a younger sibling enrolling in the same program if an older sibling wins a lottery. If an older sibling wins a lottery, a younger sibling's probability to be enrolled in the program where an older sibling has *applied* increases by 3.2 percentage points. This translates into a 98.7% increase relative to the control mean (which is reported in parentheses throughout the rest of the paper). A younger sibling is 1.2 percentage points more likely to be enrolled in the program where an older sibling is *enrolled* (21.6%). Based on the predictions derived earlier, these results indicate that the information effect is stronger than the preference effect.

In other words, younger siblings, whose older siblings lost a lottery, are less likely to enroll in a lottery program and more likely to follow their older sibling's study program choice. This indicates that common educational preferences formed before the realization of the lottery outcome are not strong in influencing sibling spillovers. Siblings are following each other due to the information sharing about the study program.

Siblings spillovers on university enrollment

Alternatively, one can argue that siblings prefer to study with each other regardless of the information about the study program. If younger siblings derive utility from studying with their siblings, there would be a positive effect of sibling spillovers on whether siblings study together. To check this explanation, I estimate the effect of an older sibling's study choice on whether a younger sibling is enrolled in the same university as an older sibling in Table 6. There is a precisely estimated zero effect of sibling spillovers on studying in the same university. This indicates

Table 5: Reduced–form estimates of the effect of older sibling's study choice of younger sibling's study choice

Outcome	YS is enrolled in	
	lottery program	same program
OS wins a lottery	0.032***	0.012***
s.e.	(0.003)	(0.004)
Control mean	0.032	0.056
Effect size	98.7%	21.6%
N	27642	27642

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. *p<0.1; **p<0.05; ***p<0.01

that siblings do not simply follow their older siblings because they prefer to study together, but due to the information–sharing mechanism. There is also no effect on whether a younger sibling is enrolled in a university, which suggests that younger siblings do not simply imitate their older sibling's study decisions, but rather get valuable information about their field of study.

Table 6: Reduced–form estimates of the effect of older sibling's study choice of younger sibling's enrollment in a university and in the same university as their older siblings

Outcome	YS is enrolled in	
	university	same university
OS wins a lottery	0.004	0.005
s.e.	(0.006)	(0.006)
Control mean	0.810	0.207
Effect size	0.5%	2.4%
N	27642	27642

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. *p<0.1; **p<0.05; ***p<0.01

6.2 Heterogeneity results

This section presents a heterogeneity analysis of sibling spillovers in study choice by sex, age difference, ability, migration background, and parental income.

By age difference

Table 8 presents the results by the age difference between siblings. Column (1) includes siblings who are less than 2 years apart, Column (2) – between 2 and 4 years, Column (3) – between 4 and 6 years, and Column (4) – who are more than 6 years apart.

The strongest effect of an older sibling study choice on whether a younger sibling is enrolled in the program where an older sibling has *applied* is among siblings with an age difference between 4 and 6 years. The smallest effect is for siblings with an age difference of fewer than 2 years. These results indicate that information accumulated during the study period by an older sibling increases with the study experience. This is intuitive since siblings who are 2 years older than their younger siblings can provide information only about the application process, and about the first year of studying. The higher the age difference is, the more information an older sibling can provide about the program. If siblings are more than 6 years apart, the information effect is decreasing as older siblings can provide less relevant information to their younger siblings.

Overall, sibling spillovers are stronger for siblings who are further apart in age. These findings are in line with the results by Altmejd et al. (2021) for the US, where they find that siblings born 5 or more years apart have stronger spillovers. The fact that effects grow with the age difference and do not disappear if there is more than 6 years difference between siblings again indicates that sibling spillovers are not driven by the desire of younger siblings to study together with older siblings, but rather by the information sharing about study experience of older siblings.

By sex composition

Table 9 presents the results of sibling spillovers by the sex composition of sibling pairs. Columns (1) and (2) show results for same–sex pairs of brothers and sisters, while Columns (3) and (4) show results for opposite–sex pairs: older brothers –

younger sisters and older sisters – younger brothers. Overall, there is not much systematic heterogeneity of sibling spillovers among siblings of the same and different sexes. One interpretation of the absence of heterogeneous effects by sex composition is that there is no sibling rivalry effect. Joensen and Nielsen (2018) find stronger siblings spillover effects in high school among brothers and they interpret these findings as evidence of sibling rivalry. My findings suggest that sibling completion is not a driving force of sibling spillovers since there are no heterogeneous effects by sex compositions. In other words, siblings appear to behave rationally in their study choice decisions by taking their sibling's experience regardless of sex or competition.

By ability

Table 10 shows the results by the ability of older siblings. As a proxy of ability, I use the lottery category of an older sibling. Note the the higher the high school GPA of an older sibling, the higher the lottery category. Column (1) shows the results of sibling spillovers by older siblings whose GPA is between 8.0 and 8.5 (lottery category B), Column (2) – GPA is between 7.5 and 8.0 (lottery category C), Column (3) – GPA is between 7.0 and 7.5 (lottery category D), Column (4) – GPA is between 6.5 and 7.0 (lottery category E), Column (5) – GPA is below 6.5 (lottery category F).

Sibling spillovers seem to be similar across older siblings of different ability levels. These findings provide suggestive evidence against siblings simply imitating their older siblings. If siblings were to imitate their older ones, there would be stronger spillover effects from higher ability older siblings as they serve as a better example of academically successful students. Hence, younger siblings would be better off imitating their more successful older siblings than less successful ones.

By migration background

Table 11 presents results by the migration background. Column (1) presents the results of sibling spillovers for siblings with a non–western background and Column (2) focuses only on siblings with no migration background.

Siblings spillovers are stronger among siblings with non–western migration backgrounds. These findings suggest that migrant families rely more heavily on the

experience of older siblings in study choice decisions due to a possible lack of information. Interestingly, sibling preferences are correlated to a higher extent among families with a migrant background. One explanation is that families with migrant backgrounds depend more heavily on their families as well. This might potentially be detrimental to children's education if educational inequality persists across generations.

By parental income

Table 12 presents results by parental income. I split the sample of sibling pairs by the median income of both parents. The information effects are identical across families with different parental backgrounds, indicating that older siblings provide valuable information to their younger siblings regardless of their family socio–economic background. On the contrary, siblings' educational preferences are not correlated within poorer families. This indicates that families with lower socio–economic backgrounds influence their children's study choices to a lesser extent, compared to families with higher SES.

In sum, the results of the main analysis and heterogeneous effects are consistent with the information–sharing mechanism. Older siblings provide information about the application process and study experience which is otherwise hard to observe. I rule out the imitation and sibling rivalry channels as a possible explanation of sibling spillovers in study choice.

6.3 Robustness checks

Table 7 presents the results of sibling spillover effects using different sample definitions. In the main analysis, I use the oldest older sibling to determine the treatment status of all younger siblings, and these are baseline results in Column (1). In Column (2) I use the younger oldest sibling, in Column (3) – only families with one older sibling participating in a lottery, in Column (4) – only families with two siblings, and in Column (5) I do not put an upper bound on the age difference between siblings. Overall, the results are robust to different sample definitions.

Table 7: Reduced–form estimates of the effect of older sibling's study choice of younger sibling's enrollment with different sample definitions

	Outcome	Oldest OS	Youngest OS	One OS	Two siblings	No age limit
1	Lottery program	0.032***	0.031***	0.029***	0.030***	0.031***
	s.e.	(0.003)	(0.003)	(0.003)	(0.006)	(0.003)
	Control mean	0.032	0.032	0.031	0.026	0.032
	Effect size	98.7%	96.4%	94.4%	115.0%	98.4%
2	Same program	0.012***	0.012***	0.010***	0.010	0.012***
	s.e.	(0.004)	(0.003)	(0.003)	(0.008)	(0.003)
	Control mean	0.056	0.056	0.055	0.059	0.055
	Effect size	21.6%	20.5%	18.3%	17.6%	22.2%
	N	27642	27642	26646	7195	28222

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. *p<0.1; **p<0.05; ***p<0.01

6.4 Simulation

The findings of the paper show that sibling spillover effects exist and they are driven mostly by information sharing among siblings. This section provides results from the simulation analysis to determine the relative role of preference and information mechanisms.

Model

I follow the analysis of the stylized model described in Section 2. Note that the preference and information mechanisms lead to the same field of study choice of younger siblings whose older siblings are lottery winners: younger siblings always choose a lottery program. Since I am interested in disentangling the relative role of preference and information mechanisms, I focus on simulating the study choice of younger siblings whose older siblings are lottery losers.

As before, younger siblings follow where their older siblings have *applied* or are *enrolled*. Under the pure preference mechanism, younger siblings choose a program where their older siblings have *applied*. Under the information mechanism, younger siblings choose a program where their older siblings are *enrolled*. Hence, the higher the role of the information mechanism, the more likely younger siblings would choose

the program their older siblings are *enrolled*.

To simulate these fields of study choice, let me introduce some notations. Let Ω_{OS} denote a set of older sibling's field of study programs. Let Ω_{YS} denote a set of younger sibling's field of study programs. Let p_i be the probability that younger siblings would follow the choice of older siblings. Then with the probability $p_i \in [0; 1]$ a younger sibling i chooses a program from a set Ω_{OS} and with the probability $1-p_i$ – from a set Ω_{YS}. In other words, p_i is the parameter determining the relative strength of the information mechanism, i.e. *information parameter*.

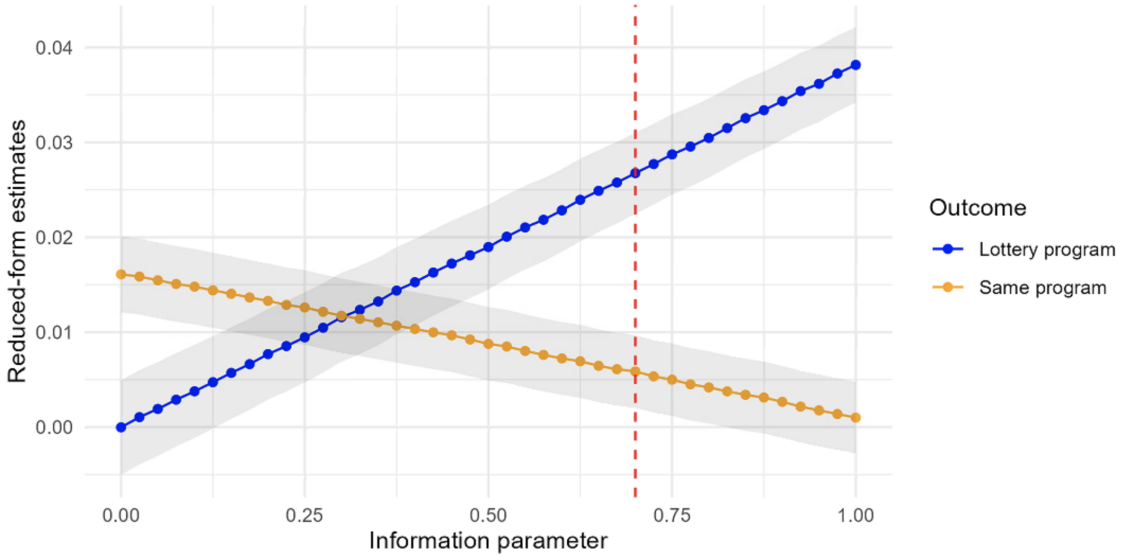
To calculate the relative strength of the information mechanism in determining sibling spillovers, I vary p_i between 0 and 1. When $p_i = 0$, the younger sibling's study choice is fully determined by the preference mechanism. When $p_i = 1$, the younger sibling's study choice is fully determined by the information mechanism. Then for each value of p_i, I draw study choices of younger siblings from actual study choices, following the procedure described above. Then, I estimate Model 1 for each draw to get reduced–form estimates of the older sibling's study choice on the younger sibling's study choice. I repeat this process 1000 times for each p_i and take an average of point estimates and 95% confidence interval.

Results

The results of the simulation analysis are in Figure 3. The blue line shows reduced–from estimates of the effect of older sibling's enrollment in a lottery program on younger sibling's enrollment in the program their older sibling have *applied*. The yellow line shows reduced–form estimates of the effect of older siblings' enrollment in a lottery program on younger siblings' enrollment in the program their older siblings are *enrolled*. The colored area is a 95% confidence interval. The red line represents the location of reduced–form estimates found in the main analysis.

As predicted by the stylized model, when the information mechanism is prevailing, i.e. $p_i = 1$, the effect of older sibling enrollment on whether a younger sibling is enrolled in the program where an older sibling has *applied* is *positive*, i.e. $\Delta_{LP} = 0.04$, while the effect on enrollment in the program where an older sibling is *enrolled* is *zero*, i.e. $\Delta_{SP} = 0$. On the other hand, when the preference mechanism is prevailing, i.e. $p_i = 0$, the effect of older sibling enrollment on whether a younger

Figure 3: Reduced–form estimates of the effect of older sibling's study choice of younger sibling's enrollment from 1000 simulations



Note: A red line shows the location of reduced–form estimates of sibling spillovers presented in the main analysis.

sibling is enrolled in the program where an older sibling has *applied* is *zero*, i.e. $\Delta_{LP} = 0$, while the effect on enrollment in the program where an older sibling is *enrolled* is *positive*, i.e. $\Delta_{LP} = 0.02$.

The reduced–form estimates of sibling spillovers found in the main analysis in Table 5 are as follows: $\Delta_{LP} = 0.032$ (se $= 0.003$) and $\Delta_{SP} = 0.012$ (se $= 0.004$). Comparing these estimates to the results of the simulation analysis, I find that 70% of sibling spillovers are due to information sharing and the remaining 30% are due to common preferences.

7 Concluding remarks

The results of the paper provide evidence that the information–sharing mechanism is prevailing in sibling spillovers. In other words, older siblings act as information providers about study options for their older siblings. Importantly, the heterogeneity of sibling spillovers in the field of study choice is not substantial so in-

formation sharing appears to be a relevant robust mechanism across different groups. These findings have relevant policy implications. Policymakers can use information campaigns to better inform students of their study options, which would lead to multiplicative effects due to spillovers.

Previous papers find no sibling spillovers on the field of study choice, but significant effects on university enrollment (Altmejd et al. (2021) in the US, Chile, Sweden, and Croatia; Aguirrea and Matta (2021) in Chile). In contrast, I find no sibling spillovers on university enrollment, while there are substantial positive effects on the field of study choice. Note that the average enrollment in universities in my sample is 80-90%, while in previous studies it is ranging from 15–18% in Chile and Croatia to 35–40% in Sweden and the United States. The pattern of these results suggests sibling spillovers depend heavily on institutional characteristics. In countries with high enrollment in universities, sibling spillovers might be less important for whether to enroll in a university (extensive margin decisions) and more important for what to study (intensive margin decisions). Hence, further research is needed to make conclusions about how sibling spillovers depend on the features of higher education systems.

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Appendix

Table 8: Reduced–form estimates of the effect of older sibling's study choice of younger sibling's study choice by the age difference

	Outcome	< 2	2-4	4-6	> 6
1	Lottery program	0.023***	0.031***	0.044***	0.035***
	s.e.	(0.007)	(0.005)	(0.010)	(0.007)
	Control mean	0.033	0.032	0.031	0.031
	Effect size	68.7%	97.8%	142.5%	112.5%
2	Same program	-0.001	0.012**	0.023**	0.018**
	s.e.	(0.008)	(0.006)	(0.010)	(0.007)
	Control mean	0.068	0.062	0.056	0.032
	Effect size	-1.8%	19.9%	41.3%	55.7%
	N	5857	10593	4702	6239

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. *p<0.1; **p<0.05; ***p<0.01

Table 9: Reduced–form estimates of the effect of older sibling's study choice of younger sibling's study choice by the sex composition

	Outcome	Male-Male	Female-Female	Male-Female	Female-Male
1	Lottery program	0.027***	0.042***	0.045***	0.022***
	s.e.	(0.008)	(0.007)	(0.009)	(0.005)
	Control mean	0.036	0.037	0.037	0.021
	Effect size	75.8%	113.9%	121.6%	105.9%
2	Same program	-0.005	0.015*	0.028**	0.011*
	s.e.	(0.009)	(0.008)	(0.009)	(0.006)
	Control mean	0.071	0.067	0.058	0.034
	Effect size	-6.6%	22.7%	48.7%	31.0%
	N	5798	7865	5621	8358

Note: Standard errors are clustered at the family level. All regressions control for non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. *p<0.1; **p<0.05; ***p<0.01

Table 10: Reduced–form estimates of the effect of older sibling's study choice of younger sibling's study choice by ability of older siblings

	Outcome	B	C	D	E	F
1	Lottery program	0.006	0.030**	0.033***	0.031***	0.030***
	s.e.	(0.021)	(0.014)	(0.007)	(0.007)	(0.005)
	Control mean	0.060	0.041	0.037	0.032	0.024
	Effect size	9.2%	73.2%	88.7%	97.4%	124.3%
2	Same program	-0.009	0.022	0.003	0.015**	0.011*
	s.e.	(0.023)	(0.015)	(0.008)	(0.007)	(0.007)
	Control mean	0.073	0.050	0.058	0.056	0.055
	Effect size	-12.8%	43.1%	5.5%	26.1%	20.8%
	N	1945	2923	6439	7175	9160

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. *p<0.1; **p<0.05; ***p<0.01

Table 11: Reduced–form estimates of the effect of older sibling's study choice of younger sibling's study choice by migration background

	Outcome	Migrant	Non-migrant
1	Lottery program	0.064***	0.030***
	s.e.	(0.017)	(0.003)
	Control mean	0.022	0.033
	Effect size	291.7%	91.1%
2	Same program	0.044**	0.011***
	s.e.	(0.019)	(0.004)
	Control mean	0.051	0.056
	Effect size	85.4%	19.7%
	N	2360	25282

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. *p<0.1; **p<0.05; ***p<0.01

Table 12: Reduced–form estimates of the effect of older sibling's study choice of younger sibling's study choice by parental income

	Outcome	< median	> median
1	Lottery program	0.021***	0.041***
	s.e.	(0.004)	(0.005)
	Control mean	0.019	0.043
	Effect size	108.3%	95.9%
2	Same program	0.002	0.020***
	s.e.	(0.005)	(0.006)
	Control mean	0.046	0.065
	Effect size	4.2%	31.0%
	N	13870	13871

Note: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for program, lottery category, year of the first lottery participation, and their interactions. *p<0.1; **p<0.05; ***p<0.01